SYDE252 - W02C – Bonus Problems

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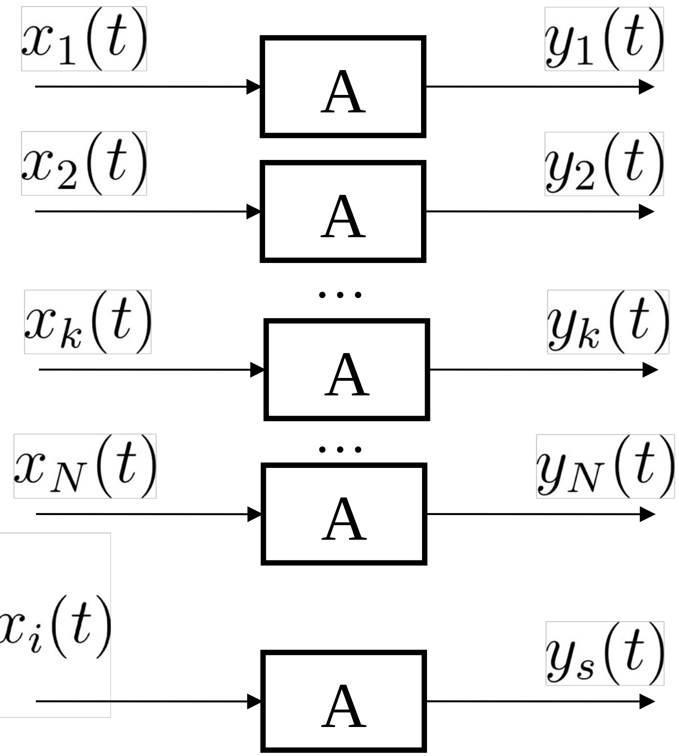
Linearity of a System



$$x_s(t) = \sum_{i=1}^{k} a_i x_i(t)$$

If and only if $y_s(t) = \sum_{i=1}^{N} a_i y_i(t)$, system A is called **Linear**.

Also called the SUPERPOSITION property!

In a bit different language:

For the input signals: **...and the system outputs:**

$$x_1(t) \longrightarrow \boxed{A} \longrightarrow y_1(t)$$

$$x_2(t) \longrightarrow \boxed{A} \longrightarrow y_2(t)$$

The system is linear if:

1. The response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$
2. The response to $ax_1(t)$ is $ay_1(t)$, where a is any complex constant

1. Is the additivity property.
2. Is the scaling (or homogeneity) property.

Check these systems for linearity...

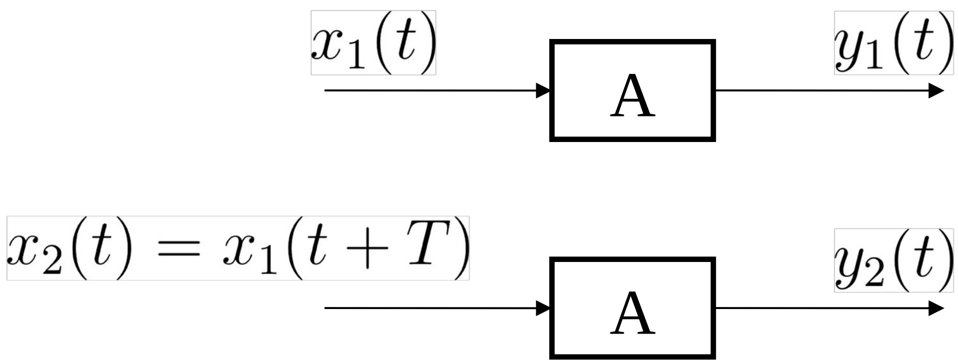
$$y(t) = x^3(t)$$

$$y[n] = 3x[n] - 2$$

$$y[n] = \mathrm{Re}(x[n])$$

Recap from last lecture

Time-invariant System



If and only if $y_2(t) = y_1(t+T), \forall t, \forall T$ system A is called **Time-invariant.**

Otherwise, the system is called time-variant.

In a bit different language:

A system is time-invariant if the behaviour and characteristics of the system are fixed over time. An example is a circuit where the resistor and capacitor values don't change over time.

For the input signals: **...and the system outputs:**

$$x(t) \longrightarrow \boxed{A} \longrightarrow y(t)$$

The system is time-invariant if:

$$x(t+t_0) \longrightarrow \boxed{A} \longrightarrow y(t+t_0)$$

A time shift in the input is seen as the same time shift in the output.

Check these systems for time-invariance...

$$y[n] = 3x[n] - 2$$

$$y(t) = x^3(t^2)$$

$$y[n] = x[n] - x[n+1]$$

So if we have both linearity and time-invariance we have a:

LINEAR
TIME
INVARIANT System

Often called an LTI system:

- If we can represent the input of an LTI system in terms of a linear combination of basic signals we can use superposition to compute the output response of the system in terms of the responses of the basic signals.
- Break down a complex problem into a bunch of simpler parts that we know the answers to then add them up!